

# RAJ POHEKAR

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## Education

**Pune Institute of Computer Technology**

*B.E in Electronics And Telecommunication Engineering*

**Nov 2022 – June 2026**

**CGPA: 8.606 / 10**

## Experience

**Research Intern - PICT, Pune**

*Elliptic Curve Blockchain Algorithm Study and Validation*

**Feb 2024 – May 2024**

**Pune, India**

- Studied **elliptic curve cryptography** for blockchain-based voting systems.
- Deduced and validated a lightweight algorithm for secure, anonymous vote recording.
- Contributed to the Springer-published paper “*Blockchain Powered Elections: A Secure and Transparent Approach*”, analyzing trade-offs between **security, scalability, and efficiency**

## Projects

**Ticket Wise - Ticket Management System** | *Node.js, Express.js, MongoDB, Inngest, Google Gemini API, JWT*

- Built an AI-powered ticket management system automating ticket categorization and skill-based moderator routing.
- Implemented **JWT-based authentication** with role-based access control (User, Moderator, Admin).
- Leveraged **Inngest** for event-driven background processing with automated email notifications.

**Vrikshami - Plant Adoption Platform** | *EJS, Node.js, Express.js, MongoDB, Razorpay, HTML, CSS*

- Built a **full-stack web platform** for NGO-based tree adoption, donations, and a **homegrown plant marketplace**, promoting sustainable community engagement.
- Designed **REST-style Express.js backend routes** to manage adoption workflows, referrals, certificate generation, and order handling.
- Integrated **Razorpay payment gateway** for secure transactions.

**ML-Based Trajectory Optimization -(Academic Project)** | *Python, Reinforcement Learning, NumPy, Matlab*

- Developed a UAV trajectory optimization system using **Reinforcement Learning (RL)** under dynamic environmental constraints.
- Implemented a **Double Deep Q-Network (DDQN)** utilizing experience replay and target networks to overcome Q-value overestimation and stabilize training.
- Conducted **data preprocessing** and simulation using **NumPy** and **Python**, evaluating model performance via cumulative reward, convergence rate, and policy efficiency metrics.
- Designed a custom reward function to balance energy efficiency, path length, and collision avoidance.

## Technical Skills

**Languages:** C++, C, Python(Basics), SQL

**Web Technologies:** Node.js, Express.js, React.js, EJS, REST APIs, JWT

**Tools:** Git, GitHub, MongoDB, Razorpay API, Gemini API

**Data Science:** NumPy, Pandas, Machine Learning, Reinforcement Learning

**CS Fundamentals:** DSA, DBMS, OS, OOP, SDLC

**Soft Skills:** Communicative, Receptive, Collaborative, Social Service

## Achievements & Leadership

- **Top 15 Finalist** – National Level Hackathon (Techathon 2.0); **NCC ‘A’ Certificate** (Bravo Grade)
- **Activity Coordinator** – PICT NSS (community outreach, NSS Camp 2025); **Contributor** – IEEE DECODE RUSH 2.0 (problem design)
- **District Cricket League Winner (2×)**; Amravati Division Trials Qualified